

❏ 1. User & Group Management

❏ User Management

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Create a user:

```
bash
```

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```
useradd username adduser username # Debian-based systems passwd username
```

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Modify a user:

```
bash
```

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`usermod -s /bin/bash username` # Change shell `usermod -aG groupname username` # Add to group `usermod -L username` # Lock user

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Delete a user:

`bash`

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`userdel username` `userdel -r username` # Also delete home directory

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View user information:

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`/etc/passwd`: User info

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`/etc/shadow`: Passwords (hashed)

- id username: UID, GID, groups
- who, whoami, w: Logged-in user info

📄 Group Management

- **Create a group:**

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```
groupadd groupname
```

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Add user to group:

bash

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```
gpasswd -a username groupname
```

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Remove user from group:

bash

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```
gpasswd -d username groupname
```

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Change user's primary group:

bash

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```
usermod -g groupname username
```

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Delete a group:

bash

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```
groupdel groupname
```

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View groups:

bash

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groups username cat /etc/group

❏ 2. Disk Management

❏ Disk Concepts

- MBR (Master Boot Record):
 - Legacy partitioning system
 - Max 4 primary partitions

- - Supports disks up to 2TB

- GPT (GUID Partition Table):**

- - Modern standard
- - Supports >128 partitions
- - Supports disks >2TB

- Filesystem Types:**

- - ext4, xfs, btrfs, FAT32, NTFS

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Common commands:

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lsblk: Lists block devices

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blkid: Displays UUIDs and filesystem info

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df -h: Shows mounted partitions

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mount, umount

☒ 3. Lab: fdisk, gdisk, LVM

☒ fdisk (MBR Partitioning)

Used for partitioning disks using MBR:

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`fdisk /dev/sdX`

- n: Create a new partition
- d: Delete a partition
- p: Print partition table
- w: Write changes

🔗 **gdisk (GPT Partitioning)**

Used for GPT partition tables:

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`gdisk /dev/sdX`

- Similar options to `fdisk`, but works with GPT

🗒 Format and Mount

bash

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`mkfs.ext4 /dev/sdX1 mount /dev/sdX1 /mnt umount /mnt`

❏ LVM (Logical Volume Manager)

LVM allows flexible disk management by abstracting physical storage.

Components:

- PV (Physical Volume): /dev/sdb1
- VG (Volume Group): Group of PVs
- LV (Logical Volume): Mountable volumes from VG

Workflow:

bash

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```
pvccreate /dev/sdb1 vgcreate my_vg /dev/sdb1 lvcreate -L 5G -n my_lv my_vg mkfs.ext4 /dev/my_vg/my_lv mount /dev/my_vg/my_lv /mnt
```

Useful Commands:

- vgs, lvs, pvs: View LVM info
- lvextend, lvreduce: Resize LVs
- vgextend, vgreduce